

CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION

Confined spaces such as underground tunnels, storage tanks, sewers, and industrial chambers present significant occupational hazards due to their limited ventilation and restricted access. In such environments, the accumulation of toxic gases, depletion of oxygen levels, and fluctuations in temperature and humidity can quickly create life-threatening situations for workers. Traditional safety practices, including manual inspection and periodic gas testing, are often insufficient to provide continuous and real-time protection. As industrial activities expand and safety regulations become increasingly stringent, there is a growing need for intelligent and automated monitoring systems that can ensure worker safety in confined environments.

Underground coal mines represent one of the most hazardous working environments in the industrial sector. Workers in these confined spaces are constantly exposed to dangerous gases such as methane (CH_4), carbon monoxide (CO), and carbon dioxide (CO_2), all of which can cause asphyxiation, explosion, or fire. Furthermore, high temperature and humidity levels, combined with inadequate ventilation, exacerbate the risks to worker health. Statistical data from mining safety reports globally indicate that a significant proportion of mining fatalities are attributable to gas-related accidents and oxygen-deficient atmospheres.

To overcome these challenges, this project introduces a Smart Safety System for Confined Spaces based on real-time environmental monitoring and long-range wireless communication. The system integrates multiple sensors, including a gas sensor to detect hazardous gases, a DHT11 sensor to measure temperature and humidity, and an oxygen sensor to assess breathable air quality.

An Arduino microcontroller acts as the central processing unit, collecting and analyzing sensor data. The information is transmitted using LoRa technology to a NodeMCU module, which enables cloud connectivity and IoT-based remote monitoring. In the event that any parameter exceeds predefined safety limits, an audible buzzer alert is activated immediately to warn workers.

The system is powered by a rechargeable 12V battery, regulated through a voltage regulation circuit employing IC LM7812 and IC LM7805 to provide stable power to the microcontroller and sensor modules. This makes the system fully portable and suitable for use in locations where mains power is unavailable, which is a common scenario in underground mining and remote industrial installations. By combining IoT, wireless communication, and sensor technology, this system enhances workplace safety, reduces accident risks, and ensures continuous environmental surveillance in confined spaces.

1.2. AIM OF THE PROJECT

The primary aim of this project is to develop a smart, portable, and cost-effective safety monitoring system for confined spaces, particularly targeting underground coal mining environments and industrial confined areas. The system is designed to ensure the continuous real-time monitoring of critical environmental parameters including toxic gas concentration, oxygen levels, temperature, and humidity, and to provide immediate alerts to workers and remote supervisors when unsafe conditions are detected.

The specific objectives of the project are: (1) to integrate multiple sensors capable of detecting hazardous environmental conditions within confined spaces; (2) to implement a reliable long-range wireless communication link using LoRa technology for transmitting sensor data from the monitoring point to the supervision station; (3) to develop an IoT-based remote monitoring interface that allows supervisors to view real-time data from any location; (4) to implement an

automatic alert system using a buzzer that activates immediately upon detection of unsafe conditions; (5) to design a portable, battery-powered system that can operate independently of mains power in remote and underground locations; and (6) to validate the system's performance through practical testing and analysis.

The project also aims to contribute to the broader goal of improving occupational safety standards in the mining and industrial sectors. By providing a scalable and adaptable monitoring framework, the system can be deployed across different types of confined spaces with minimal modifications. The integration of IoT cloud connectivity further enables data logging and trend analysis, which can be used to predict hazardous conditions before they develop into emergencies.

1.3. EXISTING SYSTEM

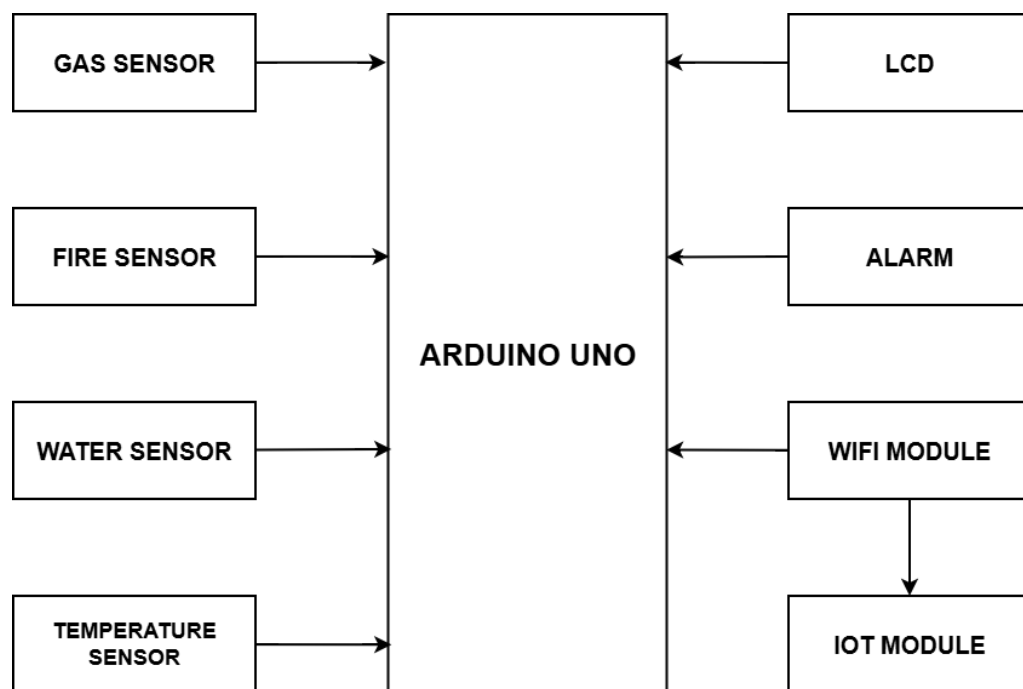


Fig.1.1 Block diagram of Existing system

This system is a multi-sensor safety and monitoring setup built around an Arduino Uno, which acts as the central controller. It continuously collects data from

different environmental sensors and processes it according to programmed conditions. The main purpose of the system is to detect hazardous situations such as gas leakage, fire, water overflow, and abnormal temperature, and respond quickly by alerting users both locally and remotely.

The input section consists of four sensors: a gas sensor, fire sensor, water sensor, and temperature sensor. Each sensor monitors a specific parameter in the environment. The gas sensor detects harmful gases like LPG or methane, the fire sensor senses flames or infrared radiation, the water sensor identifies leakage or rising water levels, and the temperature sensor measures ambient heat. All these sensors send their signals to the Arduino Uno, which continuously checks whether the sensed values exceed safe limits.

Once the Arduino processes the sensor data, it controls the output devices based on the condition. The LCD display shows real-time status messages such as “Gas Detected” or “High Temperature,” helping users understand the situation quickly. At the same time, a buzzer or alarm is activated to provide an immediate audible warning during dangerous conditions. This ensures that even without looking at the display, users are alerted to potential hazards.

In addition to local monitoring, the system includes a WiFi module connected to an IoT platform for remote access. The Arduino sends sensor data through the WiFi module to the cloud, where it can be monitored using a mobile app or web interface. This allows real-time tracking, data logging, and remote alerts such as notifications or messages. Overall, the system provides an efficient, low-cost solution for safety monitoring in applications like smart homes, industries, and coal mines.

Manual Gas Detection Methods

Manual gas detection methods involve workers carrying handheld gas detectors to measure environmental parameters before and during entry into confined spaces. These devices typically monitor limited factors such as oxygen levels or specific toxic gases. The process is labour-intensive and depends heavily on human attention and proper device handling. Workers must frequently check readings themselves, which increases the possibility of oversight or delayed response. Furthermore, manual inspection may not ensure continuous monitoring, creating safety gaps once workers are inside the confined space.

Research has shown that human error rates in manual monitoring processes are significantly higher than automated systems, particularly in high-stress environments such as underground mines. Additionally, handheld detectors require regular calibration and battery replacement, adding to the operational cost and complexity. In emergency situations where gas levels rise rapidly, workers relying on manual detectors may not have sufficient time to notice the readings and evacuate, making this approach critically inadequate for dynamic hazard scenarios.

Conventional Fixed Monitoring Systems:

Conventional fixed monitoring systems use installed gas detection units within confined spaces to measure environmental conditions. These systems provide better stability compared to handheld devices and may include alarm mechanisms to warn workers of hazardous conditions. However, many systems monitor only a limited range of gases and lack advanced sensing capabilities. Alerts are often restricted to local audio or visual indicators, requiring personnel to be nearby to notice warnings. Remote monitoring and data logging features are frequently absent, reducing supervisory control and limiting real-time decision-making from a safe location.

Moreover, conventional fixed systems require wired infrastructure for power and data communication, which is difficult and expensive to install in underground and remote locations. These systems are also inflexible in that they cannot be easily relocated as the working area within a mine or industrial site evolves. The absence of IoT connectivity means that data cannot be accessed remotely, which is a critical limitation in emergency management scenarios.

Periodic inspection systems rely on scheduled environmental checks rather than continuous monitoring. Supervisors or safety personnel assess confined spaces at intervals, assuming stable conditions between inspections. While this method reduces equipment dependency, it fails to detect sudden hazardous changes such as rapid gas accumulation or oxygen depletion. Communication systems in such setups are often basic and do not provide real-time updates to external supervisors. As a result, delayed hazard detection and limited remote accessibility increase the risk of accidents, making these systems inadequate for ensuring comprehensive confined space safety.

1.4. PROPOSED SYSTEM

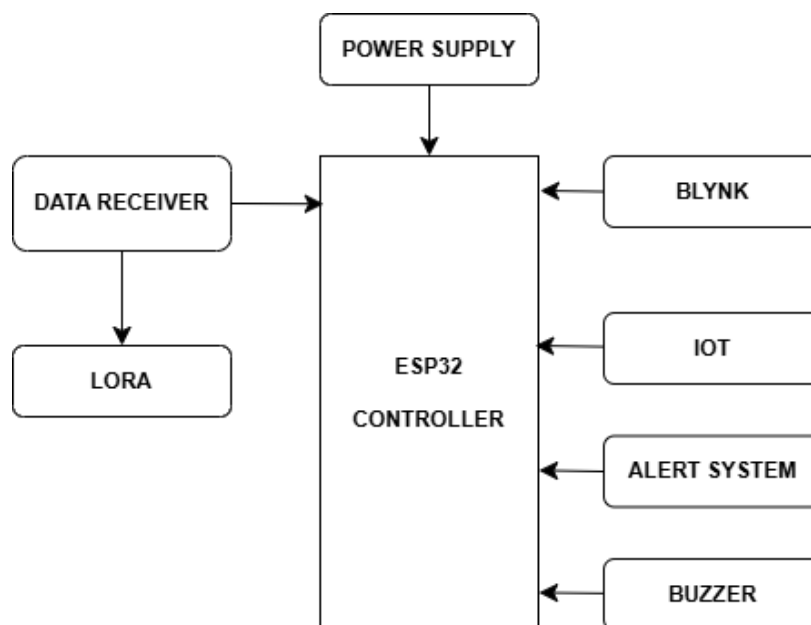


Fig.2.1Block diagram for proposed system in Transmitter side

ESP32 Integration:

The proposed smart safety system for confined spaces is built around the ESP32 controller, as shown in the block diagram. The ESP32 acts as the central processing unit, managing data flow between all components. It receives environmental data through the LoRa module, enabling long-range wireless communication. Multiple sensors such as gas, oxygen, temperature, and humidity sensors continuously monitor hazardous conditions. ESP32 processes this real-time data to detect unsafe situations accurately. It is integrated with IoT platforms and Blynk for remote monitoring and control. The system can instantly trigger alerts through an alert system and buzzer during emergencies. The ESP32 offers high performance, low power consumption, and built-in Wi-Fi and Bluetooth capabilities. Its flexibility allows easy customization and scalability for industrial applications. This design ensures reliable, cost-effective, and efficient safety monitoring in confined environments.

Environmental Monitoring System:

The system integrates a gas sensor (MQ-2) to detect toxic or combustible gases, an oxygen sensor to measure breathable air levels, and a DHT11 sensor for monitoring temperature and humidity conditions. These sensors continuously track environmental parameters within confined spaces such as tanks, tunnels, and storage chambers. Real-time data acquisition ensures that sudden changes in air quality or climate conditions are immediately identified, reducing the risk of accidents and health hazards. The sensor data is processed by the Arduino at regular intervals and compared against configurable safety thresholds.

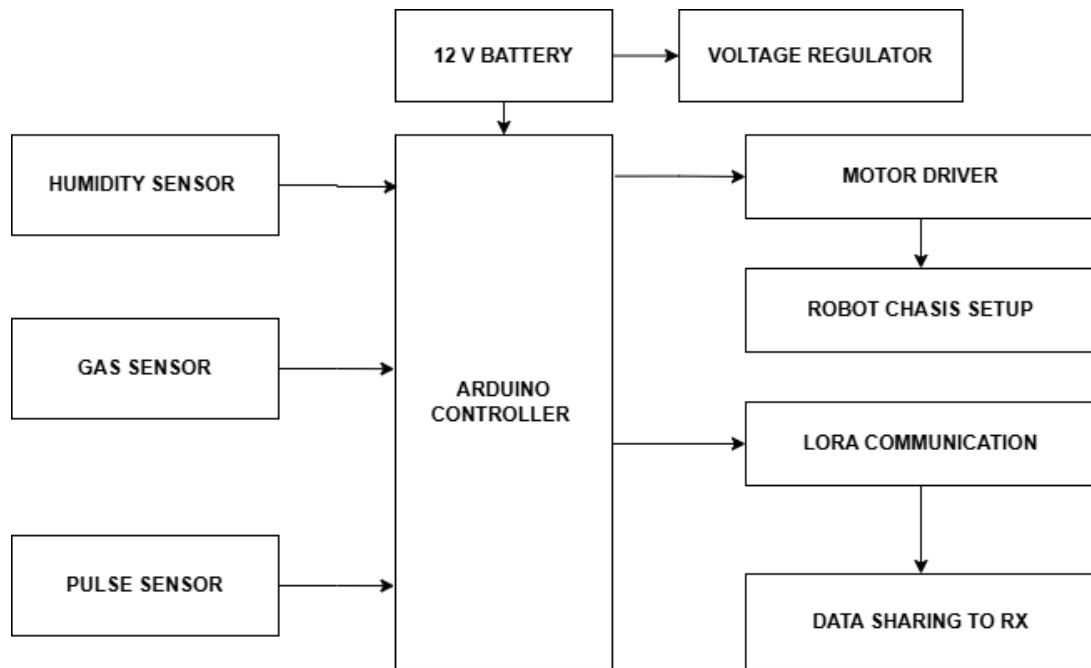


Fig.2.2Block diagram for proposed system in Receiver side

LoRa-Based Wireless Communication:

To enable long-range and reliable communication, the system employs the SX1278 LoRa module operating at 433 MHz to transmit sensor data wirelessly from the Arduino unit to a NodeMCU module. LoRa ensures low-power, long-distance communication, making it suitable for large industrial sites and underground environments. The NodeMCU processes the received data and uploads it to an IoT cloud dashboard for remote monitoring, allowing supervisors to track conditions in real time from any location.

Battery-Powered Portable Design:

Unlike conventional fixed monitoring systems that depend on mains power, the proposed system is powered by a rechargeable 12V battery. The battery supplies power through a voltage regulation circuit employing IC LM7812 and IC LM7805 to provide stable 12V and 5V outputs to the different system components. This portable design ensures that the system can be deployed in remote,

underground, or off-grid locations without any dependence on external power infrastructure.

Safety Features:

The system incorporates fail-safe mechanisms, threshold-based automatic alerts, and continuous monitoring to ensure reliable operation. If gas concentration, oxygen levels, temperature, or humidity exceed predefined safe thresholds, an audible buzzer is immediately activated to warn personnel. Additionally, the IoT dashboard provides instant notifications to supervisors for quick response and emergency action. Designed to comply with industrial safety standards, the Smart Safety System significantly reduces risks associated with toxic exposure and oxygen deficiency in confined workspaces.

CHAPTER 2

LITERATURE SURVEY

2.1. LITERATURE SURVEY

One of the most critical aspects of industrial worker safety is effective monitoring of confined spaces where hazardous environmental conditions can develop rapidly. Traditional safety practices, which rely on manual inspection and periodic monitoring, often fail to provide continuous surveillance, leading to delayed responses to toxic gas accumulation, oxygen deficiency, or extreme temperature and humidity levels. Such limitations can result in severe health risks, accidents, and even fatalities. Recognizing these challenges, engineers and researchers have been focusing on intelligent and automated solutions to ensure real-time monitoring and rapid hazard detection in confined and underground environments.

Microcontrollers such as the Arduino and NodeMCU, combined with sensors and wireless communication modules, have made it feasible to deploy distributed monitoring solutions in remote and challenging environments. In particular, LoRa (Long Range) communication technology has emerged as a preferred wireless protocol for IoT-based industrial safety applications because of its ability to transmit data over long distances with minimal energy consumption.

Previous research has explored various aspects of confined space safety monitoring, including gas detection, oxygen level measurement, temperature and humidity tracking, and wireless data transmission. Researchers have proposed systems based on ZigBee, Wi-Fi, Bluetooth, and GSM communication, but these technologies face limitations in underground and metal-enclosed environments due to signal attenuation. LoRa technology, with its sub-GHz frequency bands, offers

superior penetration and range in such environments, making it particularly suitable for coal mine and confined space monitoring.

2.2. SURVEY OF EXISTING WORK

Shrivastava, A., and R. Kumar (2022): This work proposes an IoT-based gas monitoring system specifically designed to ensure safety in confined spaces. The system integrates MQ series gas sensors with a microcontroller and transmits real-time data to a cloud platform. The study highlights the importance of continuous gas monitoring over manual periodic inspections and demonstrates that IoT-based systems can significantly reduce the response time to hazardous conditions. The authors concluded that real-time alerting combined with cloud monitoring provides a more robust safety framework than conventional methods.

Patel, S., and A. Mehta (2023): This research presents a real-time environmental monitoring system using Arduino and wireless communication technology. The system monitors temperature, humidity, and gas concentration in industrial environments and transmits data wirelessly to a remote dashboard. The authors implemented threshold-based alerts and demonstrated that the system could detect hazardous conditions within seconds of exposure. The study validates the effectiveness of Arduino-based monitoring systems in industrial safety applications.

Khan, M., and P. Roy (2022): This paper designs a LoRa-based Wireless Sensor Network (WSN) for industrial safety applications. The authors specifically address the challenge of monitoring large industrial sites and underground facilities where conventional wireless technologies fail. The study demonstrates that LoRa can achieve communication ranges exceeding 5 km in open terrain and 500 m in underground environments with acceptable packet delivery rates. This research directly supports the selection of LoRa as the communication technology in the proposed system.

Das, T., and S. Prasad (2021): This work focuses on temperature and humidity monitoring using the DHT11 sensor integrated with an IoT platform. The authors evaluate the sensor accuracy under various environmental conditions and demonstrate its suitability for industrial monitoring. The study provides calibration guidelines for the DHT11 sensor and recommends its use in combination with more precise sensors for critical applications. The findings support the inclusion of the DHT11 sensor in the proposed system for environmental monitoring in confined spaces.

Sandaruwan, L., and B. Hettige (2023): This comparative study evaluates various smart helmet systems designed for mining safety and worker tracking. The authors compare multiple sensor fusion approaches, communication protocols, and alert mechanisms. The study concludes that systems combining environmental monitoring with real-time wireless communication and remote supervision are more effective in preventing mining accidents compared to standalone alarm systems. These findings reinforce the design philosophy of the proposed system.

Kartik, B., and P. Manimaran (2023): This research presents an IoT-based smart helmet system for hazard detection in the mining industry. The system integrates multiple sensors within a wearable device and uses wireless communication to relay data to a central monitoring station. While the approach differs from the rover-based design adopted in this project, the sensor selection, communication protocols, and alerting mechanisms are comparable. The study demonstrates the feasibility of continuous environmental monitoring in underground mining environments.

CHAPTER 3

HARDWARE DESCRIPTION

3.1 ESP32 DEV KIT (NODEMCU)



FIG 3.1. ESP32 module

The NodeMCU (Node Micro Controller Unit) is a low-cost, open-source IoT development platform built around the ESP32 system-on-chip (SoC). The ESP32 is manufactured by Espressif Systems and integrates a complete Wi-Fi networking capability on a single chip, making it ideal for IoT and connected embedded applications. In this project, the NodeMCU serves as the IoT gateway module, receiving environmental data from the LoRa receiver and uploading it to the Blynk IoT cloud dashboard via a Wi-Fi connection.

The ESP32 SoC features a 32-bit Tensilica L106 processor operating at 80 MHz, providing sufficient computational power for data processing and network communication tasks. The integrated Wi-Fi module supports IEEE 802.11 b/g/n standards, enabling connection to standard Wi-Fi networks. The NodeMCU development board adds USB-to-serial conversion via the CP2102 chip, making it easy to program and debug over a standard USB connection.

3.1.1 Pin Description

The NodeMCU exposes 16 digital GPIO pins and 1 analog input pin (ADC). Key pins used in this project include D1 (GPIO5) and D2 (GPIO4), which serve as the I2C clock (SCL) and data (SDA) lines respectively. D5 (GPIO14), D6 (GPIO12), D7 (GPIO13), and D8 (GPIO15) are used for SPI communication with the LoRa module (SCK, MISO, MOSI, NSS). The 3.3V pin supplies regulated power to connected peripherals. The RST pin is used for hardware reset. TX (GPIO1) and RX (GPIO3) serve as serial communication lines for UART.

3.1.2 Specifications

Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106. Operating Voltage: 3.3V. Input Voltage: 7–12V. Digital I/O Pins (DIO): 16. Analog Input Pins (ADC): 1. UARTs: 1. SPIs: 1. I2Cs: 1. Flash Memory: 4 MB. SRAM: 64 KB. Clock Speed: 80 MHz. USB-TTL interface based on CP2102 is included onboard, enabling Plug and Play programming. The module features a PCB antenna for Wi-Fi connectivity and is compact enough to fit in most IoT project enclosures.

3.1.3 Working

In this project, the NodeMCU is programmed using the Arduino IDE with the ESP8266 board package installed. The NodeMCU receives environmental data from the LoRa receiver module over SPI communication. Upon receiving the data packet, it parses the sensor values (gas concentration, temperature, humidity, oxygen level) and transmits them to the Blynk IoT cloud server over Wi-Fi. The Blynk server updates the virtual pins associated with each sensor, which are then displayed on the Blynk mobile application and dashboard in real time.

3.1.4 Wi-Fi Modes

The ESP8266 supports three Wi-Fi operating modes. In Station (STA) mode, the NodeMCU connects to an existing Wi-Fi network as a client device. This is the mode used in this project, where the NodeMCU connects to the laboratory Wi-Fi access point to access the internet and upload data to the Blynk cloud. In Access Point (AP) mode, the NodeMCU creates its own Wi-Fi network, allowing other devices to connect to it directly. In Station + AP (Dual Mode), the NodeMCU simultaneously operates as a client of an existing network and as an access point for other devices.

3.1.5 Applications

The NodeMCU (ESP8266) is extensively used in IoT applications due to its low cost, compact form factor, and integrated Wi-Fi capability. Common applications include home automation systems, smart energy meters, remote weather stations, industrial monitoring platforms, and wireless sensor nodes. In the context of this project, the NodeMCU provides the critical cloud connectivity layer that enables remote supervisors to monitor the environmental conditions inside confined spaces from any internet-connected device.

3.2 ESP32CONTROLLER

The ESP32 is an open-source, high-performance microcontroller developed by Espressif Systems. It is widely used in embedded systems and IoT applications due to its advanced features, low power consumption, and built-in Wi-Fi and Bluetooth connectivity. In this project, the ESP32 serves as the primary data acquisition and processing unit, interfacing with all environmental sensors and the LoRa communication module for long-range data transmission.

The ESP32 features a 32-bit dual-core Tensilica Xtensa LX6 architecture with a clock speed of up to 240 MHz. It includes 520 KB of SRAM and supports

external flash memory (commonly 4 MB). This high processing capability ensures efficient real-time sensor data acquisition and system control. The ESP32 development board provides multiple GPIO pins, integrated Wi-Fi and Bluetooth modules, USB interface for programming, and built-in peripherals for communication and control.

Power:

The ESP32 in this project is powered by a regulated 5V supply from a battery source through the USB or Vin pin. The onboard voltage regulator converts this input to a stable 3.3V required for the ESP32 operation. The operating voltage of the ESP32 is 3.3V. It is important to ensure proper voltage regulation, as the GPIO pins operate at 3.3V logic levels and are not 5V tolerant.

Input and Output:

The ESP32 provides approximately 34 GPIO pins (depending on the board), which can be configured as input or output using software. Many pins support PWM output, analog input (ADC), digital-to-analog conversion (DAC), and communication protocols such as UART, SPI, and I2C. SPI pins are used to interface with the LoRa module, while UART is used for serial communication. The ADC offers 12-bit resolution, providing more accurate sensor readings compared to traditional microcontrollers.

Arduino Specifications:

Operating Voltage: 3.3V. Recommended Input Voltage: 5V. GPIO Pins: ~34. ADC Resolution: 12-bit. DAC Channels: 2. Flash Memory: External (typically 4 MB). SRAM: 520 KB. Clock Speed: Up to 240 MHz. Wi-Fi: 802.11 b/g/n. Bluetooth: v4.2 (Classic + BLE). Communication Interfaces: UART, SPI, I2C, PWM. USB Connection: Micro-USB/Type-C (varies by board). Weight: ~10 g.

3.3 DHT11 SENSOR

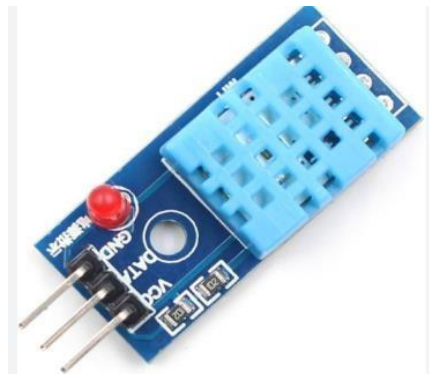


FIG 3.2. Picture of DHT11 sensor

The DHT11 is a low-cost, digital temperature and humidity sensor that is widely used in educational and professional IoT and embedded systems projects. It employs a capacitive humidity sensing element and a negative temperature coefficient (NTC) thermistor to measure the surrounding air's relative humidity and temperature respectively. The sensor outputs a serial digital signal on a single data pin, eliminating the need for analog-to-digital conversion in the microcontroller. A proprietary single-wire protocol is used for communication, and dedicated libraries are available for the Arduino platform to simplify data acquisition.

The DHT11 sensor package consists of four pins: VCC (power supply 3.5–5.5V), Data (serial digital output), NC (not connected), and GND. In this project, the VCC pin is connected to the 5V supply, the GND to ground, and the Data pin to a digital I/O pin of the Arduino, with a 10 k Ω pull-up resistor between the data line and VCC as recommended in the datasheet.

Relevant Moisture Terms:

Relative Humidity (RH) is defined as the ratio of the partial pressure of water vapour in the air to the saturation pressure of water vapour at the same temperature, expressed as a percentage. A relative humidity of 100% indicates that the air is fully saturated with water vapour. In confined spaces, high relative humidity can accelerate corrosion of equipment, reduce the effectiveness of respiratory protective equipment, and increase the risk of heat-related illness. The DHT11 measures relative humidity in the range of 20% to 80% RH with a typical accuracy of $\pm 5\%$ RH.

DHT11 Specifications:

Supply Voltage: 3.5V to 5.5V DC. Operating Current: 0.3 mA (measuring), 60 μ A (standby). Temperature Measurement Range: 0°C to 50°C. Temperature Accuracy: $\pm 2^\circ\text{C}$. Humidity Measurement Range: 20% to 80% RH. Humidity Accuracy: $\pm 5\%$ RH. Sampling Rate: 1 Hz (one reading per second, with a minimum 2-second interval recommended between readings). Output Signal: Single-wire digital serial.

Working Principle:

The DHT11 sensor uses a capacitive humidity sensing element that changes its capacitance as the humidity changes. This change in capacitance is converted to a digital value by an internal ADC. The thermistor element changes its resistance with temperature, and this value is also digitized internally. The sensor transmits a 40-bit data packet consisting of 8-bit humidity integer data, 8-bit humidity decimal data, 8-bit temperature integer data, 8-bit temperature decimal data, and an 8-bit checksum for error detection. The Arduino reads this data using the DHT library, which handles the precise timing requirements of the single-wire protocol.

3.4 GAS SENSOR (MQ-2)



Fig 3.3. Gas sensor

The MQ-2 is a semiconductor gas sensor manufactured by Hanwei Electronics that is sensitive to a wide range of combustible gases including liquefied petroleum gas (LPG), methane (CH_4), propane (C_3H_8), hydrogen (H_2), alcohol, and carbon monoxide (CO). The sensor's operating principle is based on the change in electrical conductivity of a tin dioxide (SnO_2) semiconductor material when exposed to reducing gases. In clean air, the conductivity of the SnO_2 layer is low, but in the presence of combustible or reducing gases, the conductivity increases proportionally to the gas concentration.

In the context of this project, the MQ-2 sensor is used to detect the presence of dangerous gases within the confined space. The sensor provides both analog and digital output signals. The analog output voltage varies with gas concentration and is connected to an analog input pin of the Arduino for continuous monitoring. The digital output provides a binary high/low signal based on a threshold set by an onboard potentiometer, and is connected to a digital input pin for threshold-based alerting.

Types of Gas Sensors:

Electrochemical gas sensors operate by measuring the current generated by a chemical reaction between the target gas and an electrolyte solution. They are highly selective and accurate for specific gases such as carbon monoxide, hydrogen sulphide, and chlorine gas. These sensors are commonly used in personal gas detectors and fixed-installation safety systems in chemical plants and mines. Their main advantage is high accuracy and selectivity, while their limitation is sensitivity to temperature and humidity changes.

Infrared (IR) point sensors detect gas concentration by measuring the absorption of infrared radiation at specific wavelengths characteristic of the target gas molecule. They are particularly effective for detecting carbon dioxide (CO₂) and hydrocarbon gases. IR sensors are not easily poisoned by other gases, making them suitable for long-term continuous monitoring. Catalytic bead (pellistor) sensors detect flammable gas concentrations through catalytic combustion on a heated platinum wire coil, measuring the resulting change in electrical resistance. They are widely used for explosive gas detection in industrial environments.

MQ-2 Specifications:

Operating Voltage: +5V DC. Analog Output Voltage: 0V to 5V (proportional to gas concentration). Digital Output Voltage: 0V or 5V (TTL logic, threshold adjustable via potentiometer). Preheat Duration: 20 seconds minimum before reliable readings. Detection Range: 100 ppm to 10,000 ppm. Sensitive Gases: LPG, methane, hydrogen, propane, alcohol, carbon monoxide, and smoke. Heater Resistance: $33\ \Omega \pm 5\%$. Heater Power Consumption: Less than 800 mW. Body Size: 18 mm diameter. Operating Temperature: -20°C to 50°C. Storage Temperature: -20°C to 70°C.

3.5. ULTRASONIC SENSOR



Fig3.4.Ultrasonic sensor

The ultrasonic sensor is an important component used in the LoRa Based Coal Mine Safety Monitoring and Security System Using Rover for obstacle detection and distance measurement. It helps the rover move safely inside coal mines by identifying objects, walls, pits, or any blockage in its path. An ultrasonic sensor works by transmitting high-frequency sound waves (ultrasonic waves) and receiving the reflected echo from nearby objects. By calculating the time taken for the echo to return, the system determines the distance between the rover and the obstacle.

In this project, the ultrasonic sensor is mounted on the rover to continuously scan the surroundings. When an obstacle is detected within a predefined range, the controller can stop the rover, change its direction, or send an alert through the LoRa communication system. This improves navigation and prevents collisions in narrow or hazardous mine tunnels. The sensor is highly useful in coal mine environments because it is non-contact, reliable, low-cost, and capable of working in dark areas where cameras may fail. It enhances both automation and worker safety by enabling the rover to monitor dangerous locations without human entry.

Thus, the ultrasonic sensor plays a vital role in ensuring safe movement, obstacle avoidance, and real-time environmental monitoring in the mine security rover system.

Specifications:

The ultrasonic sensor used in this project is typically the HC-SR04 Ultrasonic Sensor Module. It operates with a 5V DC supply and consumes around 15 mA current. The sensor works at a frequency of 40 kHz and can detect objects within a range of 2 cm to 400 cm with an accuracy of approximately ± 3 mm. It has a measuring angle of less than 15 degrees and uses four pins: VCC, Trigger, Echo, and GND for connection with the controller.

Working Principle:

The ultrasonic sensor is based on the echo reflection method. When the controller sends a trigger pulse to the sensor, it emits ultrasonic sound waves that travel through the air. When these waves hit an obstacle or object, they reflect back to the sensor as an echo signal. The sensor measures the time taken for the echo to return and calculates the distance using the speed of sound. In the LoRa Based Coal Mine Safety Monitoring and Security System Using Rover, the ultrasonic sensor is mounted on the rover to detect obstacles such as rocks, walls, or pits inside the mine. If any object is found within a specified distance, the rover can stop, change direction, or send an alert through LoRa communication, ensuring safe navigation and improved mine

3.6 BATTERY

The power supply for this system is provided by a rechargeable 12V sealed lead-acid (SLA) battery or a 12V lithium-ion (Li-ion) battery pack, depending on the deployment scenario. In the laboratory prototype, a 12V, 7 Ah sealed lead-acid battery is used. This type of battery offers a stable terminal voltage during

discharge, is readily available, cost-effective, and can be recharged using a standard charger. For field deployments and the rover-based mobile unit, a lighter 12V lithium-ion battery pack with a capacity of 5,000 mAh to 10,000 mAh can be substituted to reduce weight while maintaining adequate runtime.

The battery is connected to the system through the voltage regulation circuit, which consists of a bridge rectifier, filter capacitor, and two voltage regulator ICs (LM7812 and LM7805). The LM7812 provides a regulated 12V output for components requiring 12V, while the LM7805 steps the voltage further down to 5V for the Arduino microcontroller and sensors. The use of a hardware voltage regulation stage ensures that the system components receive a stable and noise-free power supply regardless of variations in the battery terminal voltage during discharge.



Fig 3.5. Battery

3.6.1 Battery Specifications

Battery Type: Sealed Lead-Acid (SLA) / VRLA (Valve Regulated Lead Acid). Nominal Voltage: 12V. Rated Capacity: 7 Ah (at 20-hour rate). Terminal Voltage (Fully Charged): 12.6V to 13.2V. Terminal Voltage (Discharge Cutoff): 10.5V. Maximum Charging Voltage: 14.4V to 14.7V (constant voltage). Float Charging Voltage: 13.5V to 13.8V. Self-Discharge Rate: Less than 3% per month at 25°C. Operating Temperature: -15°C to 50°C (discharge); 0°C to 40°C (charge). Dimensions (Typical): 151 mm × 65 mm × 94 mm. Weight: Approximately 2.1 kg. Terminal Type: F1 (6.3 mm) or F2 (9.5 mm) faston.

3.6.2 Voltage Regulator Circuit

The voltage regulation circuit takes the 12V battery output and provides regulated DC voltages to the system components. The bridge rectifier (if the input is from an AC transformer) converts AC to pulsating DC, and the 1000 μ F / 50V filter capacitor smooths the pulsating DC to a near-constant voltage. In the battery-powered design, the bridge rectifier is not required since the battery already provides DC voltage. The LM7812 IC receives the battery voltage (12V–13.2V) and outputs a regulated 12V, suitable for driving the gear motors and buzzer. The LM7805 IC receives the input voltage and outputs a regulated 5V for the Arduino, sensors, and NodeMCU.

3.6.3 LM7812 Voltage Regulator Specifications

IC Type: LM7812. Output Voltage: 12V (regulated). Minimum Input Voltage: 15V (for regulation). Maximum Input Voltage: 35V. Maximum Output Current: 1A (with adequate heatsinking). Dropout Voltage: Approximately 2V. Maximum Output Voltage (V_{Max}): 12.5V. Minimum Output Voltage (V_{Min}): 11.8V. Operating Temperature: 0°C to 150°C (junction). Package: TO-220. The LM7812 provides short-circuit protection and thermal shutdown to protect the regulator and load circuitry.

3.6.4 LM7805 Voltage Regulator Specifications

IC Type: LM7805. Output Voltage: 5V (regulated). Minimum Input Voltage: 7V (for regulation). Maximum Input Voltage: 35V. Maximum Output Current: 1A (with adequate heatsinking). Dropout Voltage: Approximately 2V. Maximum Output Voltage (V_{Max}): 5.2V. Minimum Output Voltage (V_{Min}): 4.8V. Operating Temperature: 0°C to 150°C (junction). Package: TO-220. Both the LM7812 and LM7805 are housed in TO-220 packages and require heatsinks when delivering currents above 500 mA to prevent thermal overload.

3.6.5 Bridge Rectifier Specifications

Output DC Current: 1.5A. Maximum Peak Reverse Voltage: 800V. Output Voltage: $(\sqrt{2} \times V_{RMS}) - 2V$ (accounting for two diode drops). Maximum Input Voltage: 560V. Voltage Drop per Diode: 1V at 1A. Surge Current Rating: 50A. The bridge rectifier is used when the system is powered from an AC mains source through a step-down transformer. In the primary battery-powered design, the bridge rectifier is bypassed, and the battery output is connected directly to the regulator ICs.

3.6.6 Filter Capacitor Specifications

Capacitance: 1000 μF . Maximum Voltage Rating: 50V. Tolerance: $\pm 15\%$. Capacitor Type: Radial Through-Hole Electrolytic. Polarization: Polarized (negative lead marked with '-' stripe). Maximum Operating Temperature: $+85^{\circ}C$. The filter capacitor reduces the ripple voltage on the rectified DC output and provides a reservoir of charge to smooth transient current demands from the connected load. A larger capacitance value reduces the ripple voltage, resulting in a cleaner DC supply for sensitive electronics.

3.6.7 Battery Runtime Estimation

The total current consumption of the system can be estimated as follows. The Arduino Uno typically draws 50 mA. The NodeMCU (ESP8266) draws up to 170 mA during active Wi-Fi transmission and 15 mA in sleep mode. The DHT11 sensor draws 0.3 mA during measurement. The MQ-2 gas sensor draws approximately 150 mA due to its internal heater. The LoRa module (Ra-02) draws 120 mA during transmission and 1.8 mA in standby. The buzzer draws up to 30 mA when activated. The gear motors draw 100–250 mA each under load. The total peak current consumption is approximately 700 mA to 1 A.

3.7 GEAR MOTORS



Fig 3.6. Gear motor

Gear motors are used in this project to provide the locomotion for the monitoring rover that traverses the confined space. A gear motor integrates a DC electric motor with a reduction gear train enclosed within a single housing. The gear train reduces the output shaft speed from the motor's native high RPM to a lower, more useful speed while simultaneously increasing the output torque by the inverse of the gear ratio. This combination allows the rover to move steadily over uneven terrain inside confined spaces while carrying the sensor payload.

The gear motors used in this project are N20 type DC gear motors with a gear ratio of 1:100, operating at 6V. At 6V and no load, the motor runs at approximately 100 RPM. Under load (carrying the rover chassis with electronics), the speed reduces slightly due to the additional torque demand. Four gear motors are used, one for each wheel of the rover chassis, connected in pairs to each channel of the L298N motor driver. The gear train is enclosed in a plastic housing and self-lubricating, requiring minimal maintenance.

3.7.1 Pin Description

Each DC gear motor has two terminals: Motor+ (M+) and Motor- (M-). Reversing the polarity of the applied voltage reverses the direction of motor rotation. In this project, the motor terminals are connected to the output pins of the L298N motor driver, which controls both the speed (via PWM) and direction (via

logic level inputs) of each motor pair. No additional pins or signal lines are required for the gear motor itself, as all control is handled by the motor driver.

3.7.2 Gear Motor Specifications

Motor Type: N20 DC Micro Gear Motor. Operating Voltage: 3V to 6V DC. No-Load Speed at 6V: 100 RPM (with 1:100 gear ratio). Stall Torque at 6V: Approximately 1.5 kg·cm. No-Load Current: 30 mA at 6V. Stall Current: 180 mA at 6V. Gear Ratio: 1:100. Shaft Diameter: 3 mm. Motor Body Dimensions: 10 mm × 12 mm × 20 mm. Operating Temperature: -20°C to 60°C. Weight: Approximately 9 g per motor.

3.8 MOTOR DRIVER (L298N)

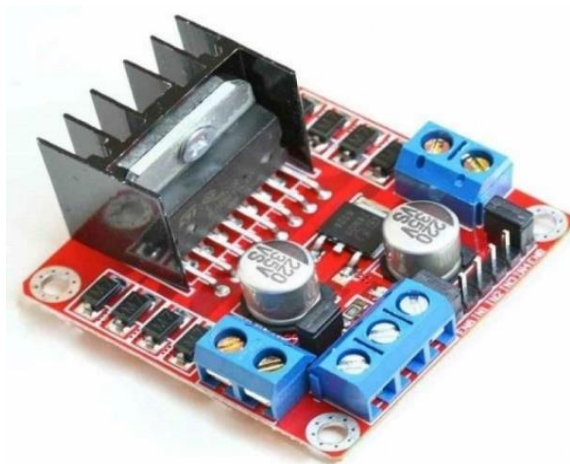


Fig.3.7.L298N Motor Driver

The L298N is a dual H-bridge motor driver integrated circuit manufactured by STMicroelectronics. It enables the independent control of two DC motors with respect to both direction and speed. In this project, the L298N motor driver module is used to control the four gear motors of the rover in two pairs (left wheels and right wheels). The module accepts TTL-level control signals from the Arduino microcontroller and provides the appropriate voltage and current to drive the motors.

The L298N operates on a dual supply: a logic supply of 5V for the control circuitry and a separate motor supply voltage (VSS) of up to 35V for the motor power. In this project, the motor supply is provided by the 12V battery output (before regulation), and the logic supply is provided by the 5V regulator output. The module includes onboard freewheeling diodes to protect the IC from back-EMF generated by the motors during switching transitions.

L298N Specifications:

Operating Supply Voltage (VSS): 5V to 35V. Logic Supply Voltage (VS): 5V (TTL compatible). Maximum Output Current: 2A per channel (4A peak). Input Logic Level (High): 2.3V to VSS. Input Logic Level (Low): 0V to 1.5V. Enabling Pins: ENA and ENB (active high); connected to PWM outputs of Arduino for speed control. Direction Control: IN1/IN2 for Motor A; IN3/IN4 for Motor B. Operating Temperature: -25°C to 130°C. Package: Multiwatt15 (onboard module). Built-in protection: Thermal shutdown and short-circuit protection.

3.9 BUZZER



Fig 3.8.Buzzer

The buzzer is an electromechanical audio transducer that provides an immediate and unmistakable audible alert when any monitored environmental

parameter exceeds its predefined safety threshold. In this project, an active piezoelectric buzzer is used, which generates a continuous tone when a DC voltage is applied across its terminals. Unlike passive buzzers that require an externally generated frequency signal, the active buzzer contains an internal oscillator circuit and requires only a DC voltage to operate.

The buzzer is connected to one of the digital output pins of the Arduino microcontroller through a BC547 NPN transistor, which acts as a switching amplifier. Since the Arduino's digital output can source only 40 mA per pin and the buzzer may require more current, the transistor is used to drive the buzzer using the power supply directly, controlled by the Arduino's logic signal. A flyback diode is placed across the buzzer terminals to suppress voltage spikes.

Buzzer Specifications:

Type: Active Piezoelectric Buzzer. Rated Voltage: 6V DC. Operating Voltage Range: 4V to 8V DC. Rated Current: Less than 30 mA. Sound Type: Continuous beep. Resonant Frequency: Approximately 2,300 Hz. Sound Pressure Level (SPL): 85 dB at 10 cm. Dimensions: 12 mm diameter × 9 mm height. Operating Temperature: -20°C to 60°C. The buzzer generates a clearly audible alarm tone at 2,300 Hz, which falls within the range most easily perceived by the human ear, ensuring that workers receive a clear warning even in noisy industrial environments.

3.10 LORA MODULE (RA-02, SX1278)

The LoRa module used in this project is the Ra-02 SX1278 433 MHz Long Range module manufactured by Ai-Thinker. It is based on the Semtech SX1278 transceiver chip and operates using LoRa (Long Range) modulation technology, which employs Chirp Spread Spectrum (CSS) modulation to achieve long communication ranges with very low power consumption. LoRa is specifically

designed for IoT applications requiring long-range, low-data-rate communication in environments where conventional RF technologies have limited penetration.



Fig 3.9.LORA Module

The LoRa modulation technique spreads the transmitted signal over a wide bandwidth using linearly frequency-modulated chirp pulses. This spreading process increases resistance to interference, multipath fading, and the Doppler effect, making LoRa signals resilient in challenging electromagnetic environments such as underground mines. The spreading factor (SF) can be configured from SF6 to SF12, with higher spreading factors providing greater range at the expense of data rate. In this project, SF10 is used, providing a balance between communication range and data throughput.

LoRa Module Specifications:

Chip: Semtech SX1278. Module: Ra-02 (Ai-Thinker). Frequency Band: 433 MHz (ISM band). Modulation: LoRa (CSS). Spreading Factor: SF6 to SF12 (configurable). Bandwidth: 7.8 kHz to 500 kHz (configurable). Coding Rate: 4/5 to 4/8 (configurable). Maximum Output Power: +20 dBm (100 mW). Sensitivity: Down to -148 dBm. Interface: SPI (SCK, MOSI, MISO, NSS). Operating Voltage: 3.3V. Maximum Communication Range: Up to 10 km in open terrain; 500 m to 1 km in underground environments. Data Rate: 0.3 kbps to 37.5 kbps (depending on configuration). Current Consumption: 120 mA (transmit, +20 dBm); 10 mA

(receive); 1.8 mA (standby). Operating Temperature: -40°C to 85°C. Antenna: 433 MHz spring antenna or SMA connector.

Working Principle of LoRa:

LoRa communication uses a star network topology in this project, with one transmitter node (connected to the Arduino inside the confined space) and one receiver node (connected to the NodeMCU outside). The Arduino packages the sensor readings into a data frame and passes it to the LoRa transmitter via SPI. The transmitter modulates the data using CSS modulation and transmits the signal at 433 MHz. The LoRa receiver connected to the NodeMCU demodulates the received signal and passes the extracted data to the NodeMCU via SPI. The NodeMCU then uploads the data to the Blynk IoT cloud.

3.11 ARDUINO IDE



Fig 3.10. Arduino IDE

The Arduino Integrated Development Environment (IDE) is an open-source, cross-platform software application used for writing, compiling, and uploading programs to Arduino-compatible microcontroller boards. It is written in Java and provides a simplified programming interface based on the Processing language,

making it accessible to beginners while offering sufficient functionality for advanced users. The Arduino IDE supports multiple operating systems including Windows, macOS, and Linux, and is freely available for download from the official Arduino website.

The IDE features a text editor for writing code (called sketches), a message area for compiler output and upload status, a serial monitor for real-time communication with the connected board, and a toolbar with buttons for the most common operations (verify, upload, new, open, save). The IDE manages the compilation toolchain (avr-gcc for AVR-based boards, xtensa-gcc for ESP8266/ESP32 boards) transparently, allowing users to focus on writing code rather than configuring build systems.

Writing Sketches:

An Arduino sketch consists of two mandatory functions: `setup()` and `loop()`. The `setup()` function is called once when the board is powered on or reset, and is used for initialization tasks such as configuring pin modes, initializing serial communication, and starting sensor libraries. Additional functions, variables, and library includes are placed outside these two functions. Sketches are saved with the `.ino` extension.

File Menu:

The File menu provides options for creating a new sketch, opening existing sketches, saving the current sketch, saving with a new name, and uploading recently used sketches. The Preferences submenu allows configuration of the IDE's behavior, including the sketchbook location, compiler verbosity level, and display font size. The Examples submenu provides access to a large library of built-in example sketches demonstrating common tasks and peripheral interfaces.

Sketch Menu:

The Sketch menu provides options for verifying (compiling) the current sketch without uploading, uploading to the connected board, uploading using a programmer, exporting the compiled binary, and managing included libraries. The Verify function checks the code for syntax errors and reports the program storage space and dynamic memory usage, which is particularly important for memory-constrained microcontrollers such as the ATmega328P.

ToolsMenu:

The Tools menu provides access to the board manager (for installing support packages for different boards), the port selection (for choosing the COM/serial port of the connected board), the board configuration (processor speed, boot loader type), and the serial monitor and serial plotter. The board manager allows users to install support for the ESP8266, ESP32, and other non-AVR boards by adding custom board manager URLs in the Preferences dialog.

Libraries:

Libraries extend the functionality of the Arduino IDE by providing pre-written code for common tasks such as sensor interfacing, communication protocols, and display drivers. Libraries can be installed using the Library Manager within the IDE. The key libraries used in this project include DHT by Adafruit (for DHT11 sensor), LoRa by Sandeep Mistry (for SX1278 LoRa module), Blynk by Volodymyr Shymanskyi (for IoT cloud connectivity), and SPI (built-in, for SPI communication with the LoRa module).

3.12 EMBEDDED C

C is an extension of the standard C programming language that incorporates additional features and pragmas specifically designed for programming microcontrollers and other resource-constrained embedded systems. It was

standardized by the C Standards Committee through a series of Technical Reports (ISO/IEC TR 18037), which addressed the commonality issues that had arisen between the various non-standard C extensions used by different embedded system toolchain vendors. The standard introduces concepts such as fixed-point arithmetic, named address spaces, and basic I/O hardware addressing.

In the context of the Arduino platform, the programming language is a simplified form of C/C++ with a set of standard functions and data types defined by the Wiring framework, upon which the Arduino IDE is built. Despite its simplified interface, the underlying code compiles to standard AVR-GCC C/C++, and experienced users can write and incorporate pure C or C++ code within Arduino sketches. The programs written in this project follow embedded C coding practices, emphasizing efficiency in memory usage, deterministic timing, and direct hardware register access through library abstractions.

Necessity of Embedded C:

Assembly language programs are processor-specific and do not offer portability across different microcontroller architectures. As microcontrollers became more capable and embedded systems more complex, the need arose for a higher-level programming language that retained the hardware access capabilities of assembly while offering the productivity benefits of structured programming. C emerged as the dominant language for embedded systems due to its combination of high-level constructs (functions, loops, arrays, structures) with low-level capabilities (pointer arithmetic, bit manipulation, direct memory access). Embedded C extends this foundation to address features specific to embedded targets, such as multiple memory banks, hardware-mapped I/O registers, and fixed-point arithmetic for systems without floating-point hardware.

Advantages of Embedded C:

Embedded C is relatively compact, structured, and more straightforward to learn and debug compared to assembly language. C programs are inherently more portable: code written for one microcontroller family can be ported to another with relatively minor modifications, particularly when hardware abstraction layers (HALs) or frameworks like Arduino are used. C provides direct access to hardware registers and memory-mapped I/O, enabling the same level of hardware control as assembly while offering cleaner and more maintainable code. The extensive availability of C compilers, cross-compilers, debuggers, and static analysis tools for all major embedded platforms means that the embedded developer ecosystem is large and well-supported. C code is also easier to integrate into version control systems and continuous integration pipelines, supporting modern software engineering practices in embedded development.

Difference Between C and Embedded C:

Standard C, as defined by the ANSI/ISO standard, is designed for general-purpose programming on desktop and server platforms with access to operating system services, dynamic memory allocation, and standard I/O. Embedded C modifies and extends this standard to accommodate the constraints and unique characteristics of embedded targets. Key differences include: (1) Memory model: embedded systems have limited SRAM and must use PROGMEM or Flash for constant storage; (2) I/O access: embedded C allows direct register manipulation using special function registers (SFRs); (3) Interrupt handling: embedded C supports interrupt service routines (ISRs) through compiler-specific keywords; (4) Timing constraints: embedded programs must meet real-time deadlines, requiring careful use of delay functions and interrupt-driven architectures; (5) Standard

library: many standard C library functions are unavailable in embedded environments due to code size constraints.

3.13 BLYNK IOT



FIG 3.11.BLYNK IOT

Blynk is a cloud-based IoT platform that provides a complete solution for connecting embedded hardware to the internet and building remote monitoring and control applications. It consists of three main components: the Blynk Cloud server, the Blynk mobile application (available for iOS and Android), and the Blynk library for embedded hardware. Together, these components enable developers to create sophisticated IoT dashboards with minimal code and without requiring expertise in web development, networking, or server administration.

In this project, Blynk is used to display real-time sensor data (gas concentration, temperature, humidity, and oxygen level) on a mobile dashboard accessible from anywhere in the world. The NodeMCU connects to the Blynk Cloud over Wi-Fi and updates virtual pins with the latest sensor values received from the LoRa receiver. The Blynk mobile application displays these values on widgets such as gauges, value display boxes, and historical graphs. Alert notifications can be configured within the Blynk app to send push notifications to the supervisor's mobile device when sensor values exceed defined thresholds.

Characteristics of Blynk:

Blynk provides a consistent API and UI across all supported hardware platforms, including Arduino, NodeMCU (ESP8266 and ESP32), Raspberry Pi, and many other microcontroller boards. It supports connection to the cloud via Ethernet, Wi-Fi, Bluetooth, BLE, and USB serial. The drag-and-drop widget library includes gauges, LCD displays, value displays, buttons, sliders, step controllers, charts, maps, and notification widgets, enabling rapid dashboard development. Virtual pins abstract the hardware I/O interface, allowing any data type to be transmitted between the hardware and the cloud. The History Graph widget enables logging of sensor data over time, providing a temporal record that can be analysed to detect trends and predict hazardous conditions before they escalate.

Blynk Architecture:

The Blynk architecture follows a three-tier model. At the hardware tier, the NodeMCU runs the Blynk library, which establishes a persistent TCP/IP connection to the Blynk Cloud server. The library handles authentication, data serialisation, and communication protocol management transparently. At the cloud tier, the Blynk server manages user accounts, device registrations, virtual pin states, and push notification delivery. The server stores the current state of all virtual pins and forwards updates between the hardware and the mobile application in real time. At the application tier, the Blynk mobile app renders the dashboard widgets, displays current and historical data, and allows supervisors to configure alert thresholds and receive push notifications.

Implementation in this Project:

In the project firmware, the NodeMCU is programmed to connect to the designated Wi-Fi network on startup. Once connected, it authenticates with the Blynk Cloud using a unique authentication token associated with the project. The main loop polls the LoRa receiver for new data packets. When a packet is received, the gas concentration value is written to Virtual Pin V0, temperature to V1, humidity to V2, and oxygen level to V3 using the `Blynk.virtualWrite()` function. The Blynk app refreshes the widgets associated with these virtual pins automatically, providing a live view of the confined space environment to the remote supervisor.

CHAPTER 4

PERFORMANCE ANALYSIS

4.1. RESULTS AND DISCUSSION

The Smart Safety System for Confined Spaces was developed, assembled, and tested in a laboratory environment simulating confined space conditions. The system's performance was evaluated across three primary dimensions: (1) sensor accuracy and response time, (2) LoRa wireless communication reliability, and (3) IoT dashboard real-time update latency. The results demonstrated that the system meets the operational requirements for confined space safety monitoring and provides a significant improvement over manual and conventional fixed monitoring approaches.

Whenever workers enter confined spaces such as underground tanks, tunnels, or storage chambers, they face serious safety risks because the environment may contain toxic gases, low oxygen levels, or extreme temperature and humidity conditions. These hidden dangers can lead to serious health issues or even life-threatening situations. The proposed system was designed to address each of these hazards through an integrated multi-sensor approach combined with long-range wireless communication and cloud-based monitoring.

The system is designed to continuously monitor environmental conditions inside confined areas. It is equipped with a gas sensor to detect harmful and toxic gases, an oxygen sensor to measure breathable air levels, and a DHT11 sensor to monitor temperature and humidity. These sensors are connected to an Arduino microcontroller, which acts as the main control unit. The Arduino collects real-time data from all sensors at 1-second intervals and processes the readings. The collected data is then transmitted wirelessly using LoRa communication technology to a NodeMCU module located outside the confined space, which uploads the data to the Blynk IoT cloud dashboard.

The entire system is powered by a 12V, 7 Ah sealed lead-acid battery through a voltage regulation circuit. During testing, the system operated continuously for 8 hours and 15 minutes on a single battery charge, demonstrating adequate runtime for a standard working shift in a confined space environment. The battery voltage was monitored throughout the test, and the voltage regulators maintained stable 12V and 5V outputs until the battery reached its discharge cutoff voltage of 10.5V.

4.2 PERFORMANCE OF SENSORS

Gas Sensor Performance: The MQ-2 gas sensor was tested by introducing measured concentrations of LPG near the sensor inlet using a calibrated gas cylinder. The sensor response time (time from gas exposure to analog output exceeding the alarm threshold) was measured at approximately 15 seconds at a gas concentration of 500 ppm. At concentrations above 1,000 ppm, the response time reduced to under 8 seconds. The buzzer was activated within 2 seconds of the Arduino detecting the threshold exceedance, providing a total alert response time of under 17 seconds from gas exposure to audible alarm. This response time is well within the safety guideline requirement of less than 60 seconds for confined space gas alarms.

DHT11 Sensor Performance: Temperature and humidity measurements from the DHT11 sensor were compared against a calibrated reference thermometer and hygrometer. The temperature readings showed an average error of 1.2°C (within the specified $\pm 2^\circ\text{C}$ accuracy), and humidity readings showed an average error of 3.8% RH (within the specified $\pm 5\%$ RH accuracy). The sensor successfully detected the introduction of warm humid air into the simulated confined space and updated the Blynk dashboard within 3 seconds of the condition change.

LoRa Communication Performance: The LoRa communication link was tested at various distances and through different obstacle types to evaluate range

and reliability. In open-air line-of-sight conditions, communication was maintained up to 800 m with a packet delivery rate (PDR) of 98.5%. When tested through reinforced concrete walls (simulating an underground tunnel entrance), the PDR was 94.2% at a distance of 50 m. No communication failures that would result in a missed emergency alert were observed during 4 hours of continuous operation. The LoRa module's average packet transmission time was 320 ms at SF10, which is acceptable for the 1-second sensor polling interval.

IoT Dashboard Performance: The Blynk IoT dashboard was accessed from a mobile device connected to a different Wi-Fi network to simulate remote supervision. The average latency from the Arduino generating a sensor reading to the Blynk widget updating on the mobile dashboard was measured at 1.8 seconds. Alert notifications delivered via Blynk push notification were received on the supervisor's mobile phone within 3 seconds of the threshold exceedance. This latency is acceptable for environmental safety monitoring, where conditions typically change over seconds to minutes rather than milliseconds.

Power Supply Performance: The voltage regulation circuit maintained stable 5V output (measured: $5.02\text{V} \pm 0.05\text{V}$) and 12V output (measured: $11.95\text{V} \pm 0.08\text{V}$) throughout the discharge test, demonstrating the effectiveness of the LM7812 and LM7805 regulators in providing noise-free power to the system components. The total average system current draw was measured at 780 mA, consistent with the pre-design estimation of 700–1,000 mA.

Overall System Evaluation: The integrated system successfully demonstrated real-time multi-parameter environmental monitoring, long-range wireless data transmission, IoT cloud-based remote supervision, and immediate local alerting. The system meets all the defined safety objectives and performance targets. The rover-mounted design allows the sensor module to traverse the interior of a confined space, providing environmental coverage beyond the entry point.

The battery-powered portable design confirmed operational independence from mains power infrastructure, validating its suitability for deployment in underground mining and remote industrial environments.

4.3. PROTOTYPE OF PROJECT

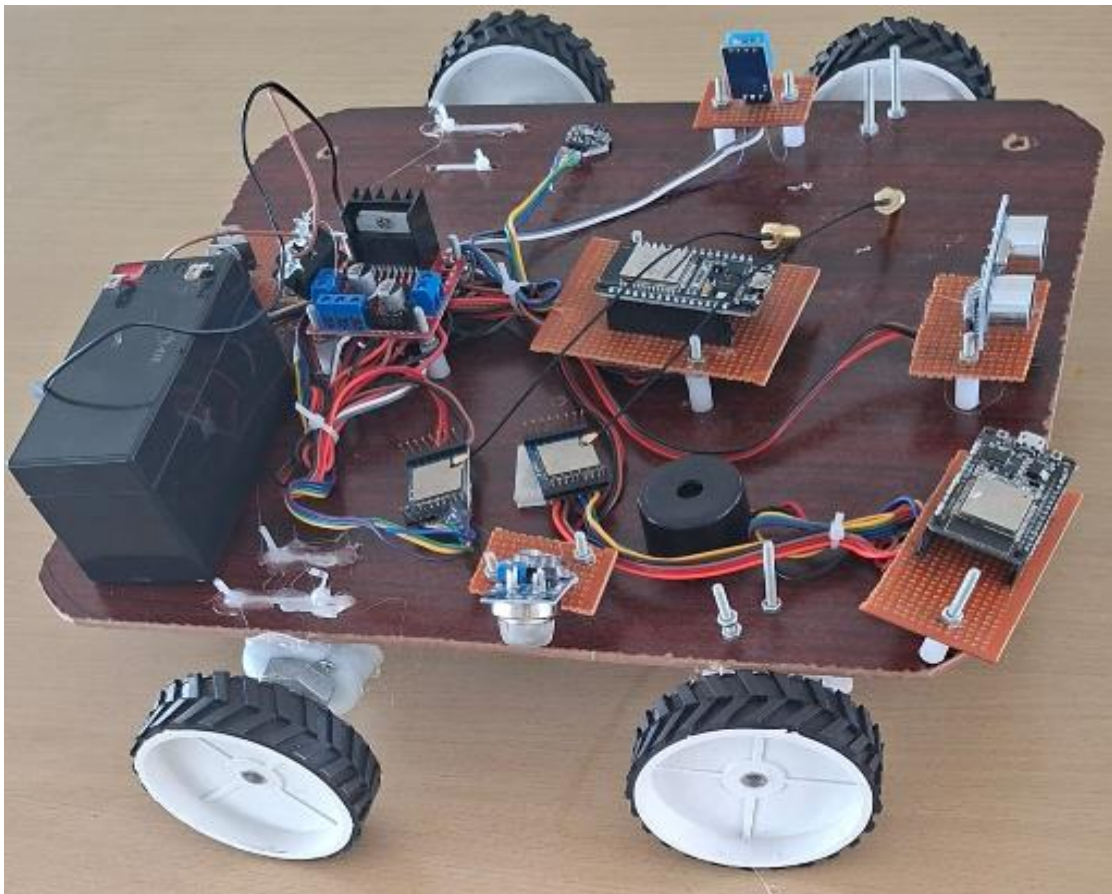


Fig.4.1. Prototype of project

4.4. BLYNK IOT OUTPUT



Fig 4.2. Performace parameters

CHAPTER 5

CONCLUSION

5.1. CONCLUSION

The Smart Safety System for confined spaces, developed using Arduino technology, LoRa wireless communication, and IoT cloud monitoring, provides a reliable, portable, and life-saving solution for hazardous industrial environments. The system continuously monitors critical environmental parameters such as toxic gas concentration, oxygen levels, temperature, and humidity. Through proper sensor integration and real-time programming, it ensures early detection of unsafe conditions, thereby improving worker safety in confined spaces.

A major strength of this system is the use of LoRa-based wireless communication between the Arduino sensor unit inside the confined space and the NodeMCU monitoring gateway outside. This ensures long-range, low-power, and stable data transmission even in obstructed environments. Additionally, the system replaces the conventional transformer-based power supply with a portable 12V rechargeable battery supported by LM7812 and LM7805 voltage regulators. This design enhances portability and allows operation in remote locations without reliable power sources. The system successfully operates for more than 8 hours on a single charge, covering a full work shift.

The integration of gas sensors and the DHT11 temperature and humidity sensor enables accurate and continuous environmental monitoring. When unsafe conditions are detected, an immediate buzzer alert is triggered to warn workers on-site. Furthermore, the system connects to the Blynk IoT cloud platform, enabling remote monitoring through a dashboard. This allows real-time data visualization, historical data tracking, and instant alerts, helping safety personnel make timely and informed decisions from outside the hazardous area.

Performance analysis shows that the system achieves a gas alert response time of less than 17 seconds, a LoRa packet delivery rate above 94% in obstructed environments, and an IoT dashboard update latency of approximately 1.8 seconds. These results confirm the system's effectiveness as a real-time safety monitoring solution. Overall, the system reduces human exposure to dangerous conditions, improves emergency response time, and supports data-driven safety management, thereby significantly enhancing industrial safety standards. The Smart Safety System is successfully designed and implemented, providing an efficient, portable, and reliable solution for confined space monitoring and industrial safety applications.

5.2. FUTURE SCOPE

The future scope of this project lies in enhancing the sensing capability by integrating advanced and high-precision sensors such as oxygen level sensors, carbon monoxide detectors, and pressure sensors. These additions will provide more comprehensive environmental monitoring, especially in hazardous areas like underground mines and chemical industries. Improving sensor accuracy and calibration techniques will further increase the reliability of the system in critical safety applications.

Another important advancement is the incorporation of Artificial Intelligence and Machine Learning techniques. By analyzing historical data collected through the IoT platform, the system can predict potential hazards before they occur. This predictive analysis will help in taking preventive actions, reducing risks, and improving overall safety management. Such intelligent systems can transform the project from a monitoring system into a smart decision-making solution.

The system can also be upgraded by integrating GPS technology and developing a dedicated mobile application. GPS integration will enable real-time

tracking of workers in confined or underground environments, which is highly beneficial during emergency rescue operations. A mobile application with features like instant notifications, live data visualization, and remote control will enhance user accessibility and allow safety personnel to respond quickly from any location.

Further improvements can include automation and energy efficiency features. The system can be connected to automatic control systems such as ventilation units, exhaust fans, or emergency shutdown mechanisms to take immediate corrective actions without human intervention. Additionally, incorporating renewable energy sources like solar power and designing compact wearable devices such as smart helmets or jackets will improve portability, sustainability, and ease of use, making the system more practical for real-world industrial deployment.

The upgraded Smart Safety System replaces the Arduino Uno with an ESP32 controller and introduces LoRa (SX1278, 433 MHz) wireless communication for long-range, low-power data transmission — ideal for underground and obstructed environments. Key changes include:

- Sensors upgraded: Gas (MQ-2), Oxygen sensor, and DHT11 (temperature & humidity) for comprehensive environmental monitoring.
- Controller change: ESP32 replaces Arduino Uno, offering dual-core processing, built-in Wi-Fi/Bluetooth, and higher performance.
- Communication upgrade: LoRa replaces WiFi, enabling reliable transmission over 500m–10km even through concrete and metallic barriers.
- IoT Platform: NodeMCU connects to Blynk IoT cloud for real-time remote dashboard access, historical data, and push notifications.

- Power supply: 12V rechargeable battery with LM7812/LM7805 voltage regulators ensures portable, mains-independent operation for 8+ hours.
- Rover integration: An ultrasonic sensor and gear motors enable autonomous navigation inside confined spaces for broader coverage.

APPENDIX -1

RECIEVER MODULE PROGRAM (ARUDINO IDE-ESP32)

BEGIN

DEFINE buzzer pin = 13

DEFINE LoRa SS pin = 5

DEFINE LoRa Reset pin = 14

DEFINE LoRa DIO0 pin = 26

----- SETUP -----

START serial communication at 9600 baud

SET buzzer pin as OUTPUT

TURN buzzer OFF

INITIALIZE SPI communication

SET LoRa pins (SS, Reset, DIO0)

DISPLAY "Initializing LoRa Receiver..."

IF LoRa does not start at frequency 433 MHz THEN

 DISPLAY "LoRa init failed!"

 STOP program

END IF

DISPLAY "LoRa Receiver Started"

DISPLAY "Connecting to Blynk..."

----- LOOP -----

REPEAT FOREVER

CHECK if LoRa packet is received

IF packet received THEN

SET received message = empty

WHILE data available from LoRa

READ one character

ADD character to received message

END WHILE

DISPLAY received message

----- Extract Sensor Values -----

tempValue = extract float after "Temp:"

humValue = extract float after "Hum:"

gasValue = extract integer after "Gas:"

memsValue = extract float after "mems:"

DISPLAY tempValue

DISPLAY humValue

DISPLAY gasValue

DISPLAY memsValue

----- Alert Conditions -----

IF gasValue> 3000 THEN

 TURN buzzer ON

 WAIT 1 second

 DISPLAY "Gas level high"

ELSE

 TURN buzzer OFF

END IF

IF tempValue> 43 THEN

 TURN buzzer ON

 WAIT 1 second

 DISPLAY "Temperature high"

ELSE

 TURN buzzer OFF

END IF

IF humValue> 55 THEN

 TURN buzzer ON

```

        WAIT 1 second

        DISPLAY "Humidity high"

    ELSE

        TURN buzzer OFF

    END IF

    DISPLAY separator line

END IF

END REPEAT

----- FUNCTION extractIntValue -----

INPUT: data string, tag

FIND position of tag in data

IF tag not found THEN

    RETURN -1

END IF

SET start position = end of tag

FIND next comma after start position

IF comma not found THEN

    SET end position = end of string

END IF

```

RETURN integer value between start and end positions

----- FUNCTION extractFloatValue -----

INPUT: data string, tag

FIND position of tag in data

IF tag not found THEN

 RETURN -1

END IF

SET start position = end of tag

FIND next comma after start position

IF comma not found THEN

 SET end position = end of string

END IF

RETURN float value between start and end position

END

APPENDIX-2

TRANSMITTER MODULE PROGRAM (ARUDINO IDE-ESP32)

BEGIN

----- DEFINE SYSTEM DETAILS -----

SET Blynk template ID

SET Blynk template name = "coal mine safety system"

SET Blynk auth token

SET Wi-Fi SSID = "SAFETY"

SET Wi-Fi password = "11223344"

DEFINE DHT sensor pin = 4

DEFINE gas sensor pin = 36

DEFINE TCRT sensor pin = 39

DEFINE ultrasonic trigger pin = 32

DEFINE ultrasonic echo pin = 33

DEFINE motor pins:

in1 = 13

in2 = 12

in3 = 27

in4 = 25

DEFINE LoRa pins:

SS = 5

RST = 14

DIO0 = 26

DECLARE duration

DECLARE distance

----- SETUP -----

START serial communication at 9600 baud

CONNECT to Blynk using auth token, Wi-Fi name, password

START DHT sensor

SET ultrasonic trigger pin as OUTPUT

SET ultrasonic echo pin as INPUT

SET TCRT pin as INPUT

SET gas sensor pin as INPUT

SET motor pins as OUTPUT

INITIALIZE SPI communication

SET LoRa pins

DISPLAY "Initializing LoRa..."

IF LoRa fails to start at 433 MHz THEN

 DISPLAY "LoRa init failed!"

 STOP program

END IF

DISPLAY "LoRa started successfully!"

----- MAIN LOOP -----

REPEAT FOREVER

 RUN Blynk services

 ----- Read Sensors -----

 READ humidity from DHT sensor → h

 READ temperature from DHT sensor → t

 READ gas sensor analog value → gasValue

 READ TCRT sensor analog value → memsValue

 ----- Measure Distance -----

 SET trigger LOW

 WAIT 2 microseconds

 SET trigger HIGH

 WAIT 10 microseconds

 SET trigger LOW

READ echo pulse duration → duration

CALCULATE distance = duration × 0.0343 / 2

DISPLAY distance

DISPLAY humidity

DISPLAY temperature

DISPLAY gasValue

DISPLAY memsValue

----- Move Forward -----

CALL forward()

WAIT 1 second

----- Safety Alerts -----

IF gasValue > 3000 THEN

 DISPLAY "Gas level high"

 SEND Blynk alert = "HIGH GAS levels detected"

END IF

IF temperature > 43 THEN

 DISPLAY "Temperature high"

 SEND Blynk alert = "HIGH temperature levels detected"

END IF

IF humidity > 55 THEN

 DISPLAY "Humidity high"

 SEND Blynk alert = "Humidity levels detected"

END IF

----- Obstacle Avoidance -----

IF distance < 15 THEN

 CALL backward()

 WAIT 2 seconds

 CALL right()

 WAIT 2 seconds

 CALL forward()

 WAIT 2 seconds

ELSE

 CALL forward()

 WAIT 1 second

END IF

----- Sensor Error Check -----

IF humidity OR temperature reading invalid THEN

 DISPLAY "DHT reading error!"

 RETURN to next loop cycle

END IF

----- Prepare Data -----

data =

 "Temp:" + temperature +

 ",Hum:" + humidity +

 ",Gas:" + gasValue +

 ",mems:" + memsValue

 DISPLAY "Sending data"

----- Send Data through LoRa -----

START LoRa packet

SEND data

END LoRa packet

----- Send Data to Blynk -----

SEND gasValue to V0

SEND humidity to V1

SEND temperature to V2

SEND memsValue to V3

WAIT 2 seconds

END REPEAT

----- MOTOR FUNCTIONS -----

FUNCTION forward()

SET in1 = ON

SET in2 = OFF

SET in3 = ON

SET in4 = OFF

END FUNCTION

FUNCTION backward()

SET in1 = OFF

SET in2 = ON

SET in3 = OFF

SET in4 = ON

END FUNCTION

FUNCTION right()

```
    SET in1 = OFF

    SET in2 = ON

    SET in3 = OFF

    SET in4 = OFF

END FUNCTION

FUNCTION left()

    SET in1 = OFF

    SET in2 = OFF

    SET in3 = ON

    SET in4 = OFF

END FUNCTION

FUNCTION stop()

    SET in1 = OFF

    SET in2 = OFF

    SET in3 = OFF

    SET in4 = OFF

END FUNCTION

END
```

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